The Realistic Warehouse World

The two logistics layouts and the four `dataset_allsensor_realistic` recordings, drawn from Chapter 3

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Purpose of this document. Chapter 3 describes the realistic warehouse family in its methodology section on simulation environments (`chapters/chapter_3_realistic_worlds.tex`) and lists its recordings in the dataset registry. This document gathers that material into one place for the reader of the Experiment 3 results: what the two layouts are, how their traffic moves, what the static twins control, what each of the four `dataset_allsensor_realistic` recordings contains, and which limits apply when they are scored. Nothing here is new evidence; every figure is the chapter's own, and the numbers come from the generator's tables and the bags.

1 Why the family exists

The earlier simulation worlds are controlled stimuli, not warehouses. Their props are spread over the floor, confined to the wall region, or shuttled back and forth along fixed segments, and their static arrangement differs in geometry from their dynamic ones, so a static-versus-dynamic comparison on them is confounded by the arrangement. The realistic family was generated to remove both objections: the layout follows the warehouse-design references collected under `docs/references/warehouse/`, whose common prescription is a chain of zones (receiving, storage, picking, packing, dispatch) served by cross-aisles with the fast-moving goods nearest the docks; and each dynamic world has a static twin that shares every footprint and differs only in whether four vehicles move. The family was implemented after the 25 September 2026 reporting cutoff (6 October) and is scored in the realistic-world half of Experiment 3.

2 The two layouts

Both canonical flow shapes of the design literature are provided, written by `script/make_realistic_world.py` from parametric tables and never edited by hand (figures R.1 and R.2).

- **U-shape flow** (`-layout u`): receiving and dispatch docks on the same north wall with the management office between them, pick faces nearest the docks, rack storage in north–south aisles, picking and packing east, block storage west, charging, empty pallets and the general office along the south wall.
- **I-shape (through) flow** (`-layout i`): receiving docks on the north wall and dispatch docks on the south wall, storage in the middle, pick faces between storage and dispatch, packing beside dispatch, block storage west, both offices on the east wall, so goods cross the building in one direction.

Everything that is not the object arrangement is held fixed: the 41.28 m shell, the plain (`nofloortexture`) floor, lights, physics configuration and robot spawn. The one change to the shell is a regenerated wall carrying dock-door openings, 3.6 m by 4.5 m; a closed door is a roll-up

panel, an open one shows a docked 13.6 m trailer and a yard slab so that a camera looking out sees a trailer interior and tarmac rather than sky. Aisles are 3.0 m between zones and 2.8 m between rack rows; pallet pitches are 3.2–4.0 m, widened from the stock 2.4 m so that a loaded pallet jack has the 2.6 m it needs to turn. Table R.1 lists the zone contents; figure R.3 shows the baked occupancy truth of each world.

Table R.1: Zones of the two worlds and the number of object instances each holds, counted from the generator's tables. "Inside" excludes the dock doors, trailers and yard. The four carriers are the pallet jacks the traffic plugin drives; the people are actors, not objects.

Zone	Contents	U-shape	I-shape
Receiving staging	pallet loads in two rows, one parked jack	11	17
Dispatch staging	pallet loads in two rows, one parked jack	7	11
Pick faces	east–west shelves with totes in front	13	8
Storage, medium rotation	three double rack rows	30	24
Picking	three double rack rows	18	18
Packing	four desks, totes, bin	8	8
Block storage	floor-stacked pallets, one empty cell per job	13	6
Management office	two desks, bin	3	3
General office	desks, bin	4	3
MHE charging	parked pallet jacks, bin	5	4
Empty pallets	stacks	2	2
Carriers (traffic)	pallet jacks on jobs	4	4
People on foot (actors)	order pickers and dock checkers	4	4
Inside the walls		118	108
Dock doors, trailers, yard		8	13
Total object instances		126	121
Baked occupancy [% of cells]		20.7	19.1

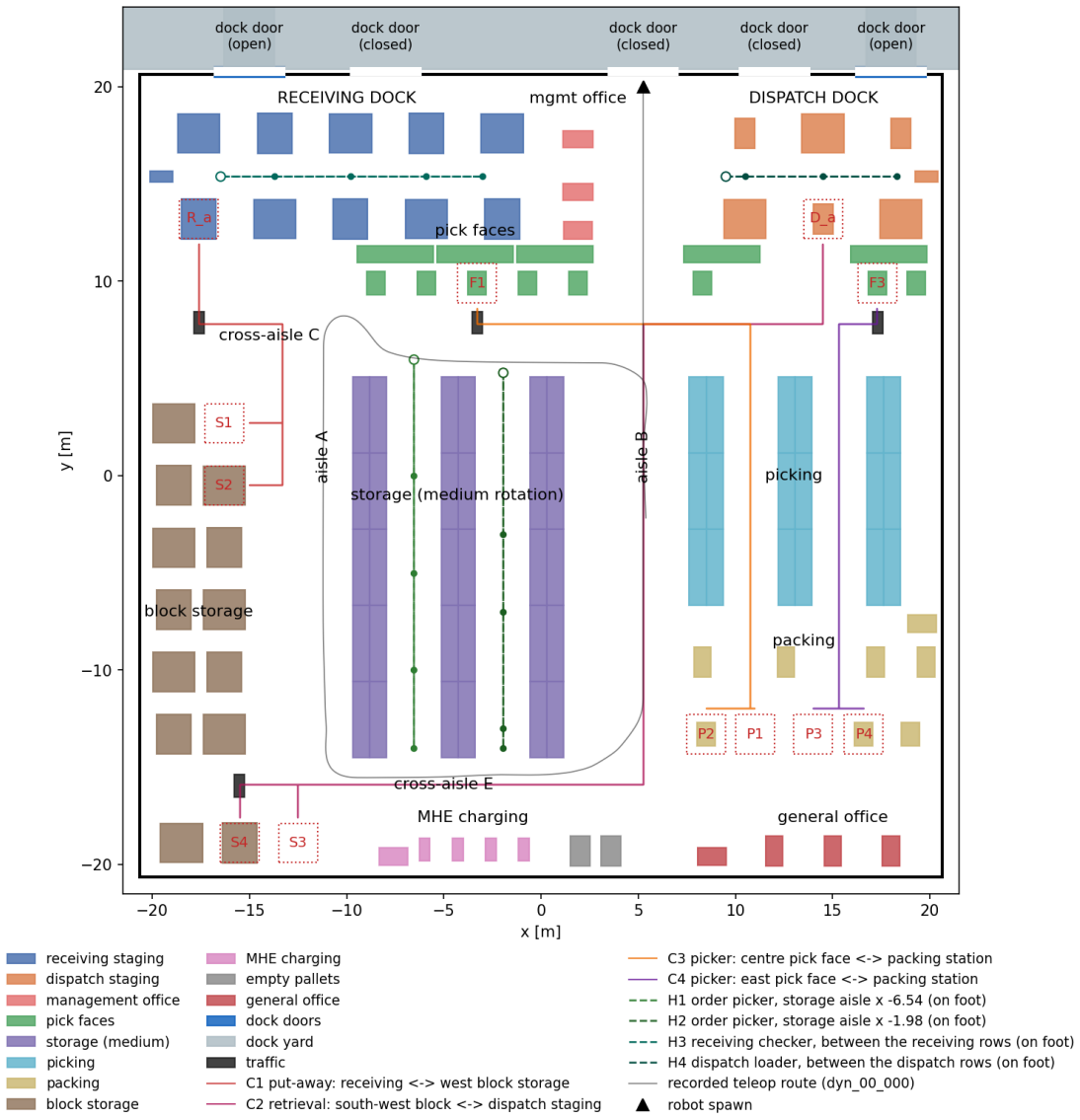


Figure R.1: Plan of the U-shape world, drawn by the generator from the tables that write the world file. Filled rectangles are object footprints coloured by zone; dotted red boxes are the exchange slots the carrier jobs pick from and drop on; coloured polylines are the four carrier routes; dashed green lines are the routes of the four people on foot, a dot at each stop and an open circle at the start; the grey trace is the ground-truth route of `dataset_dynamic_nofloortexture_00_000`, which aisles A and B and cross-aisles C and E were placed to keep drivable. The band above the north wall is the dock yard with two docked trailers.

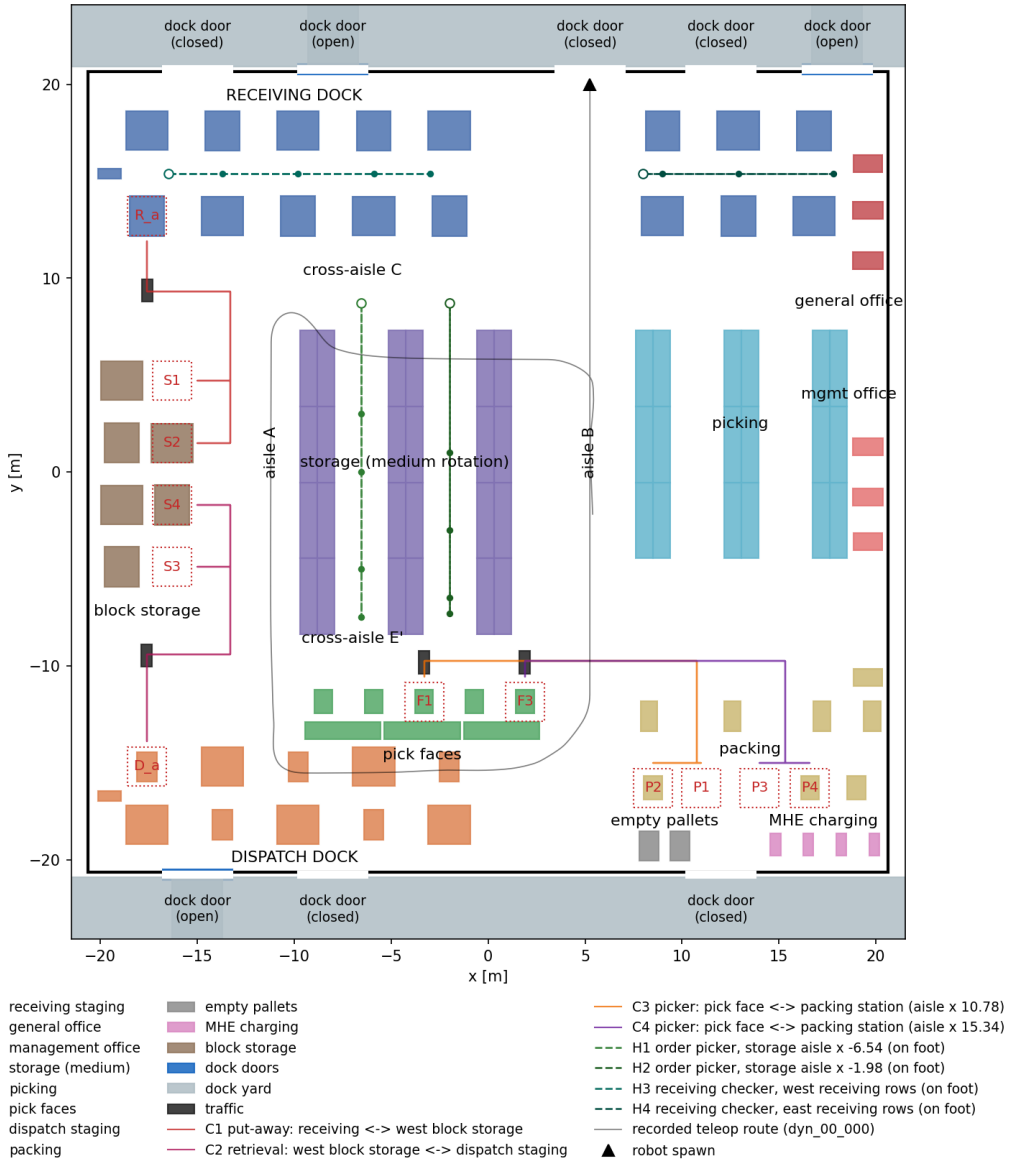
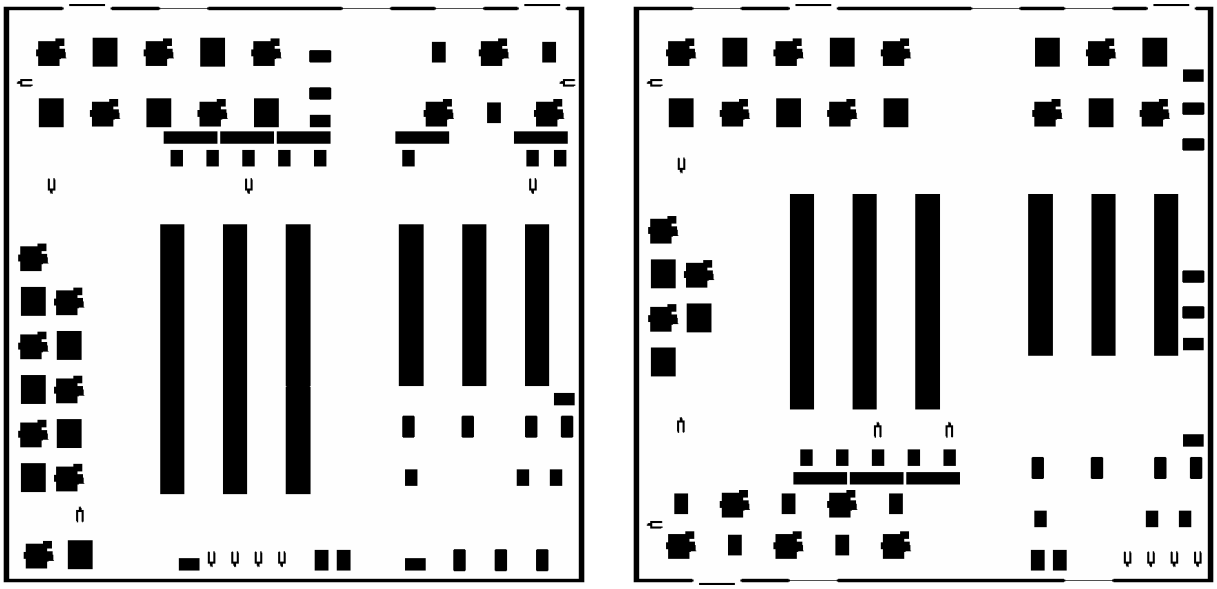


Figure R.2: Plan of the I-shape world, same conventions. Receiving docks on the north wall, dispatch docks on the south wall; the recorded teleop route (grey) crosses the storage racks here and is not drivable in this layout, which is why the I-shape recordings use a new route.



(a) U-shape, 20.7 % occupied

(b) I-shape, 19.1 % occupied

Figure R.3: Baked occupancy truth of the two worlds: 42.2 m square at 0.05 m per cell, origin $(-21.1, -21.1)$ m. Carriers and loads are baked at their $t = 0$ pose, where the static twin holds them permanently. Closed dock doors are the one-cell segments in the wall; open doors are the gaps.

3 Traffic

Carriers on jobs. The moving objects are four pallet jacks driven by a world-level Gazebo system, `gz_kinematic_trajectory/LogisticsScheduler.cc`. A carrier drives to a load, takes it onto its forks, carries it through the aisles, sets it down on a target slot and goes for the next: two jacks move pallets between receiving, block storage and dispatch, two move totes between the pick faces and the packing desks. Each job is a six-move rotation over two loads and three slots that returns to its starting state, so it loops indefinitely and no drop lands on an occupied slot. Carriers move at 1.2–1.3 m s^{-1}, turn in place at 1.5 rad s^{-1}, spend 4–5 s on each pick or drop, carry the load 1.3 m ahead of the body, enter slots head-on and leave in reverse. Before a world is written the generator proves it: every static pair at least 0.25 m apart, every job's slot bookkeeping simulated, and the swept box of carrier plus load checked on every leg and turn against every object not in its own job; the tightest clearance is 0.10 m at a packing-desk turn in each world.

People on foot. Each world carries four walking people, Gazebo actors on the stock Fuel skin with a walk clip between stops and a stand clip at them: two order pickers, one per storage aisle, who stop at bays for six seconds and turn at the aisle ends, and two dock checkers who walk the lane between the staging rows. Walking speed is 1.1–1.2 m s^{-1} with staggered starts; a 0.7 m person box is swept along every leg against the statics (at least 0.25 m), the carrier corridors and the other people, with tightest clearances of 0.59 m to a static and 0.34 m to a carrier sweep. The pickers turn at the aisle mouths, which puts them in view of a robot driving the cross-aisle.

What the traffic is not. Everything moves kinematically: the vehicles cost no physics and give no collision response, and two carriers crossing one intersection at the same moment pass through each other. Actors have no collision geometry, the occupancy bake does not see them, and the static twins omit them. One limitation matters for scoring: an actor's animated pose is computed on the rendering side and is not written back to the entity state, so `/world/default/pose/info` reports every person at its initial pose throughout a recording. Their positions can only be reconstructed from the world file and the

clock (`script/people_script.py`); a detector ground truth for them is not yet implemented, so Experiment 2 cannot score the people and Experiment 3 sees them only as unlabelled motion.

4 Static twins

For each layout a static world, `worlds/small_warehouse_static/small_warehouse_static_realistic_nofloortexture.world` and its `..._01.world` counterpart, is derived by `script/make_static_world.py`: every object is marked static and the traffic plugin removed, nothing else changed. This is the methodological point of the family. A realistic world and its twin share every footprint, so a comparison between a recording in one and a recording in the other isolates the moving vehicles. The comparison is not perfectly clean in practice: the two recordings in a pair follow routes driven by hand, so route and traffic are not separated on the U layout (section 5).

5 The four recordings

Four `dataset_allsensor_realistic` recordings were made between 9 and 14 October 2026, one per layout and twin, each carrying the full sensor set (stereo images, IMU, joint states, wheel odometry, `/clock`, `/ground_truth/odometry` and the Gazebo pose stream). Table R.2 summarises them from the bags; figure R.4 shows each layout with its recorded route.

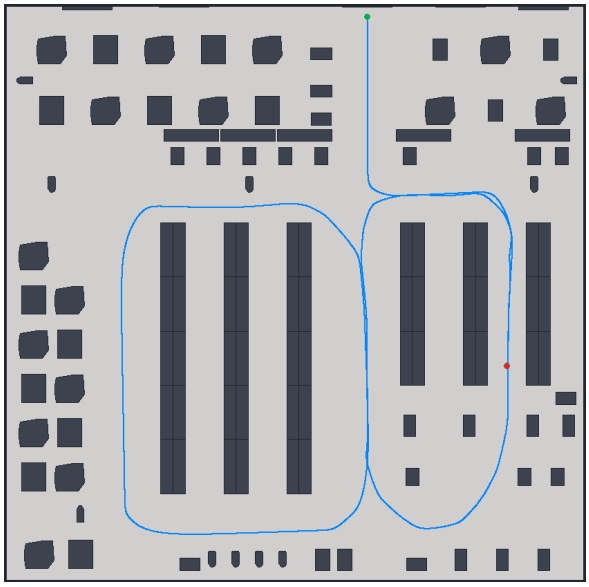
Table R.2: The four recordings (`dataset_allsensor_realistic_000` to `_003`), read from the bags (`script/render_dataset_scene.py`). "Objects" counts obstacle instances whose collision geometry intersects the robot-height band, excluding the wall ring; "Route" is the ground-truth path length; "Truth samples" is the number of ground-truth odometry messages.

Recording	Layout	Dynamics	Objects	Route [m]	Sim span [s]	Truth samples
`realistic_000`	U	static twin	125	171.3	125.5	6278
`realistic_001`	I	static twin	119	133.0	96.9	4847
`realistic_002`	U	traffic	125	102.7	79.5	3975
`realistic_003`	I	traffic	119	158.0	117.0	5852

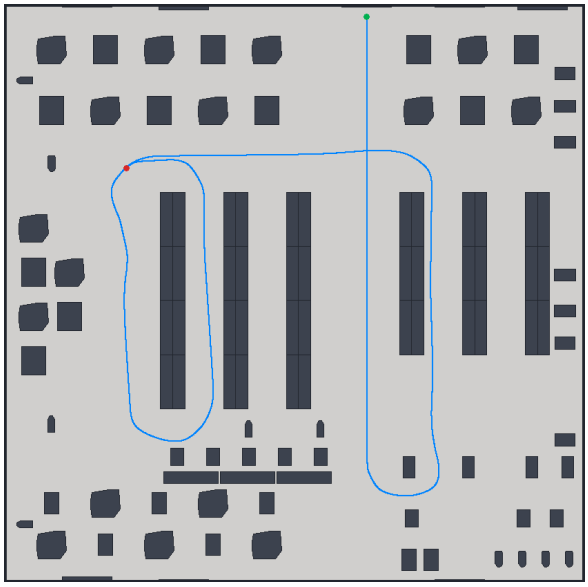
The U-shape routes are not the same in the two U recordings. The static twin's route is 171 m over 126 s; the traffic recording's is 103 m over 80 s, so on this layout a difference between the two recordings mixes route with traffic. The I-shape pair is closer in length (133 m and 158 m) but is also hand-driven. The recorded routes keep to the aisles the generator reserved: on the U layout aisles A ($x \approx -11.2$ m) and B ($x = 5.25$ m, the dock lane the robot spawns in) and cross-aisles C and E, which the original teleop route clears by at least 0.69 m; on the I layout the original route crosses the racks, so both I recordings follow a new one.

Traffic actually met. The Gazebo pose stream of the two traffic recordings names the models that moved: on `realistic_002` the four carriers and six carried loads, on `realistic_003` the four carriers and their loads likewise; the static twins contain no moving model. From those poses, the ground-truth heading and the shelf footprints, a carrier or load is in the camera's view (within 20 m, within $\pm 45°$, line of sight clear) for a median 37 % of the tracked time of the traffic recordings; the people are absent from the stream and are not counted (`script/extract_traffic.py`, Experiment 3 insight document).

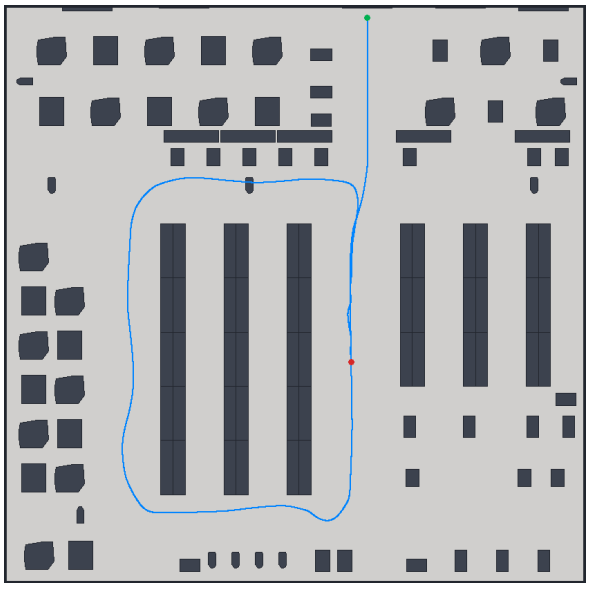
The first recording of the family. `dataset_map_realistic` (8 October 2026, 126.9 s simulated, 151.1 m route at a median 1.38 m s^{-1}) preceded the four and is the only simulation recording close to the



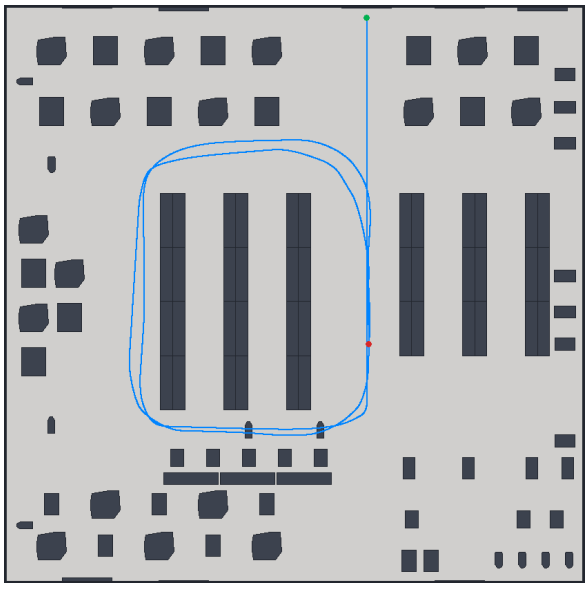
`realistic_000`: U-shape, static twin



`realistic_001`: I-shape, static twin



`realistic_002`: U-shape, traffic



`realistic_003`: I-shape, traffic

Figure R.4: Each recording as a scene: obstacle footprints at robot height (convex hulls of the collision meshes), the ground-truth route in blue from the green start to the red end. Moving carriers are drawn at their $t = 0$ pose; the people are actors without collision geometry and do not appear.

1.5 m s^{-1} of the problem statement. It was recorded without a run manifest; the bag's model inventory identifies the U-shape world with every model stationary, which is consistent with the static twin and with the dynamic world loaded without its traffic plugin. It is used for mapping, not in the Experiment 3 realistic slot.

6 Limits that apply when the recordings are scored

- **Kinematic traffic.** The vehicles and people have no physics and no collision response; a robot driving into one passes through it. The visual effect on the cameras is the one being tested; the dynamic consequence is not simulated.
- **People unlabelled.** Their poses are not in the bag, their detector truth is not implemented, and the detector weights do not include a person class. Every traffic statistic in Experiment 3 is a lower bound on the moments something moving was in the image.
- **Route and traffic not separated.** One hand-driven recording per cell; on the U layout the routes differ in length by two thirds.
- **Plain floor.** The family exists only in the `nofloortexture` condition; the bay-line floor of the earlier arrangements was not generated for it.
- **Ground-truth reference point.** `/ground_truth/odometry` reports the robot model origin; both visual estimators report the IMU, 0.338 m ahead and 0.192 m above it. Scoring moves the ground truth to the IMU point before alignment (22 October 2026); the scene figures draw the model-origin route.

[Visual verification, carried over from Chapter 3] The generator-side checks are complete, and the first Gazebo session on 6 October 2026 found and fixed one defect the checks could not see. Three things remain to be confirmed by eye rather than by the checker: that loads ride on the fork side of the jack, the shading of the generated door, trailer and yard meshes, and the open or closed state of each door. Until then the worlds are implemented in geometry and partial in visual verification.

7 Where the recordings are used

The four recordings form the realistic-world half of Experiment 3 (`insights/experiment_3/`, Section 4.3.1 of the chapter): the unfiltered baselines of both estimators on all four, the class and motion-masked arms on the two traffic recordings, and the timeline and insight analyses (`insights/experiment_3_insight/`). Their ground truth is extracted to `output/output_vins/simulation/dataset_allsensor_realistic_00*/ground_truth.csv`; the bags themselves are held on the external drive, not in the workspace.